Theo Usher

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EDUCATION

Columbia University, *Bachelor of Science, Mechanical Engineering* New York, NY; Sep 2020 – May 2024

GPA 4.07/4.0, *TBP* Engineering Honor Society, Mechanical Engineering Certificate of Merit, Magna Cum Laude

Non-Standard Courses: Managing Tech Innovation, Robotics Studio, Human-Centered Design, Public Speaking

WORK EXPERIENCE

Boeing, *Space Electronics and Mechanical Product Design Engineer* Los Angeles, CA; August 2024 – Present

- Design, integrate, and support manufacturing of electronics systems and hardware
- Develop unconventional PCB mounting solutions to meet electrical and spatial constraints
- Design hardware and software tools to improve production and evaluation efficiency
- Supported adjacent team in integrating and mounting multiple PCBs to meet a critical deadline
- Responsible Engineering Authority for Preliminary Reviews, assessing flight hardware for build quality
- Acted as production support lead, coordinating engineers and reviewing millions of dollars' worth of hardware

Airobotics Drones, *Design Engineering Intern* Tel Aviv, Israel; May 2023 – Jul 2023

- Worked on a 5-person team to build the Iron Drone autonomous, drone-catching UAV
- Analyzed, tested, and optimized UAV's net launcher for range and reliable interception
- Designed, prototyped, and constructed a new industrial drone ground station
- Optimized drone ground station design, cutting assembly time from 2 weeks to 2 days
- Engineering contributions used in Iron Drone systems sold globally for $25M+ and exhibited by the U.S. DOT

Terabase Energy, *Mechanical Engineering Intern* Davis, CA; May 2022 – Aug 2022

- Collaborated on and field-tested robotic vehicles to transport and install solar panels in utility-scale projects
- Designed a solar panel installation mechanism that more than doubled installation range
- Built a panel stabilization system that replaced high-cost linear actuators with low-cost, durable mechanisms
- Engineering contributions used in automated solar installation factories that have installed 40+ MW of panels

Columbia Bartending Agency, *Bartender* New York, NY; Dec 2022 – June 2024

LEADERSHIP & ENGINEERING CLUBS

Columbia Space Initiative, *Rockets Mission Lead* Sep 2020 – Jun 2024

- Led a 50-person team to research, design, and build a hybrid rocket to reach 30,000 ft with a scientific payload
- Organized meetings, goals, and project timelines, allowing for rapid testing of innovative designs
- Coordinated design and integration meetings, fostering cross-functional collaboration and creative ideas
- Cultivated and communicated with industry partners to raise over $15,000 in sponsorships
- Designed, manufactured, tested, and integrated the rocket electronics and recovery systems
- Led Rockets' first successful flight and recovery, improved engine performance by 10x, and doubled team size
- Built a talented team that later became the first US student-led group to launch a liquid-oxygen hybrid rocket

ENGINEERING DESIGN PROJECTS

Automated Robotic Linkage – Machine Design Sep 2023 – Dec 2023

- Worked on a team to design, build, and control a complex linkage mechanism to rapidly press arcade buttons
- Designed a unique, cable-driven mechanism unlike any our professor had seen, greatly reducing inertia

Bipedal Walking Robot Jan 2022 – May 2022

- Designed, manufactured, and programmed a bipedal robot using SolidWorks, 3D printing, and a Raspberry Pi
- Achieved the first and fastest walking robot in a class of significantly more experienced engineers

SKILLS

Technical Skills: CAD (SolidWorks, Creo, Fusion), PCBA Design (Xpedition, Eagle), MATLAB, Excel, Python, C++, FEA, GD&T, CAM, DFM, DFA, Product Design, Human-Centered Design, Root Cause Analysis, Production Support

Non-Technical Skills: Project Management, Public Speaking, Presentations, Collaboration, Communication